

P2_v2: does the Theorem-M hazard induction transfer to the block-model amplitudes $A_{\pm}$?

research programme P2, second atom, 2026-09-26. Not reviewed. Result: NO (with evidence), item (C) of P2 stays OPEN.

Task. P2_note.tex (Section ?? there) leaves OPEN the non-vanishing of the two block-model amplitudes $A_{\pm} = \frac{1}{2}(H_e \pm H_o)$ for a dense set of odd N , which is the one thing that would close item (C) (accumulation of zeros of B at every point of the small arc $\ell \leq 0$, via Rouché). The brief asked whether the hazard induction of P1e_lemma.tex (Theorem M, $Z_N \neq 0$ on $a/N \in [\frac{1}{5}, \frac{4}{5}]$) adapts, since $A_{\pm}$ are “the same kind of object” as Z_N : finite sums with near-cancellation. This note checks that structurally and numerically. Labels as in P2_note.tex/P1e_lemma.tex; PROVED, VERIFIED (numerics), HEURISTIC, OPEN.

1 The exact formulas (extracted from P2_note.tex §??)

For N odd, $\zeta = e(a/N)$, $c = -2(1 - \zeta)$, $s = c^{N/2}$ (either branch):

$$H_e = \sum_{\rho=0}^{(N-1)/2} \frac{c^{\rho} \zeta^{\rho^2}}{(\zeta; \zeta)_{2\rho}}, \quad H_o = \frac{1}{s} \sum_{\kappa=(N+1)/2}^{N-1} \frac{c^{\kappa} \zeta^{\kappa^2}}{(\zeta; \zeta)_{2\kappa-N}}, \quad A_{\pm} = \frac{1}{2}(H_e \pm H_o),$$

where $(\zeta; \zeta)_m = \prod_{j=1}^m (1 - \zeta^j)$. These are *finite* sums, $O(N)$ terms each, evaluated exactly (no truncation, no analytic continuation) by P2_heads.py/P2v2_scan.py.

2 Does Z_N ’s structural fact hold for $A_{\pm}$? NO

The hazard induction of Theorem M rests on one structural fact, stated at the top of P1e (Lemma R): Z_N is the $N \rightarrow \infty$, $a/N \rightarrow p/q$ (q fixed) limit of an object built from an analytic expansion in a separate small parameter, and the limit is literally $(1 - w_q)^{-1/2q} Z_q(p/q)$ up to an error that $\rightarrow 0$. That renormalisation is what lets the induction climb: Z_N near a rational p/q is controlled by the already-known Z_q at the smaller level q , with the gap made up by the size of the denominator (the “hazard” bookkeeping of Lemma H, purely a Diophantine-approximation argument on $\|mx\|$ for the frequency index m of an infinite convergent series).

$A_{\pm}$ have no such expansion parameter: H_e, H_o are closed finite sums over a *block index* ρ , not a Fourier/frequency index m of a convergent series in q -adic u_0 . There is no small parameter to expand in and no obvious sense in which $A_{\pm}(a/N)$ for large N should be “close to” a smaller-level object $A_{\pm}(p/q)$ or $Z_q(p/q)$. I checked this directly: does $A_{\pm}(a/N)$ converge as $a/N \rightarrow p/q$ (here $p/q = \frac{1}{13}$) along good rational approximants a/N , $N \rightarrow \infty$?

a/N	N	$ A_+ $	$ A_- $
$1/13 = 0.076923$	13	1.460	1.458
$2/27 = 0.074074$	27	2.134	0.560
$4/53 = 0.075472$	53	2.261	0.967
$6/79 = 0.075949$	79	1.379	2.278
$10/131 = 0.076336$	131	2.576	2.443
$20/261 = 0.076628$	261	5.490	5.875
$40/521 = 0.076775$	521	30.36	30.43
$80/1041 = 0.076849$	1041	874.6	874.6

(P2v2_renorm.py, mpmath 40 digits.) A_+, A_- **do not converge; they grow, roughly like e^{cN} , as $a/N \rightarrow 1/13$ with $N \rightarrow \infty$.** This is the opposite of what Lemma R gives for Z_N (a finite limit). So the one fact that makes the hazard induction work — a renormalisation onto a fixed lower-level object — fails outright for $A_{\pm}$: there is nothing to induct *down to*. VERIFIED (this table; also consistent with P2_note.tex’s own table where $|A_{\pm}|$ at $\frac{1}{16}, \frac{1}{20}, \frac{1}{24}$ already grows with N at fixed a : 2.31, 2.85, 3.43).

3 Why: the sum is spread, not edge-dominated

The hazard bookkeeping of Lemma H needs the sum to be dominated by rare “bad” terms (those m with $\|mx\|$ small, i.e. close to a low-height rational), with the rest of the terms geometrically small and summable — that is what lets a single nearby seed j/n carry the whole estimate. I checked whether H_e ’s terms have this shape: is $|c^\rho \zeta^{\rho^2} / (\zeta; \zeta)_{2\rho}|$ dominated by one or two ρ (e.g. ρ near $N/2$, where $2\rho \rightarrow N-1$ and $(\zeta; \zeta)_{2\rho}$ picks up a factor $1 - \zeta^{N-1} = 1 - \zeta^{-1}$ close to 0 exactly because a/N is small)?

At $a = 10, N = 131$ ($a/N \approx \frac{1}{13}$; P2v2_terms.py), the ten largest terms of H_e by magnitude occur at $\rho = 7, 1, 8, 14, 0, 2, 15, 9, 20, 21$, with magnitudes 2.30, 2.17, 1.93, 1.56, 1.0, 0.95, 0.75, 0.62, 0.57, 0.56 — spread over $\rho \in [0, 21]$ (out of 65 terms total), no single term or short run of terms dominating. This is the signature of a genuine oscillatory/Weyl-sum cancellation (comparable to a finite Gauss sum), not of a Diophantine-tail phenomenon. The hazard machinery has nothing to grab onto here: there is no analogue of “the deepest hazard” because closeness of a/N to a rational does not concentrate the sum on a few terms the way it concentrates Z_N ’s frequency- m tail.

4 Verdict

The hazard induction technique does not transfer to $A_{\pm}$. Two independent reasons, both checked:

- (i) No renormalisation limit exists to induct on (Section 2): $A_{\pm}(a/N)$ diverges, rather than converging to a fixed lower-level object, as $a/N \rightarrow p/q, N \rightarrow \infty$. Theorem M’s whole architecture (base case at small N + step relating level N to a smaller level q via a nearby seed) has no target for the step.
- (ii) The near-cancellation in H_e, H_o is spread over $O(N)$ terms of comparable size (Section 3), not concentrated by Diophantine proximity to low-height rationals the way Z_N ’s frequency sum is. So even a hazard-*style* argument built from scratch (not literally reusing P1e’s lemmas) would need a different mechanism: this looks like a Gauss-sum / theta-function non-vanishing problem (want: no odd- N character sum of this shape vanishes, for a dense set of N), which is a different literature and a different set of tools (stationary phase / Weil-type bounds on incomplete Gauss sums, or exact evaluation via the finite theta transformation) than continued-fraction hazard bookkeeping.

- (iii) A further disanalogy, noted but not separately checked in depth: Theorem M’s regime is $a/N \in [\frac{1}{5}, \frac{4}{5}]$ ($\ell \geq 0.855$, ζ “generic”, far from 1), and P1e’s own Remark (golden threshold) says the method provably cannot reach $\ell < 0.48$ without removing the loss factor Λ (an OPEN structured-moment induction). P2’s window is $a/N \in (0, 0.0804)$ ($\ell \leq 0$), i.e. ζ close to 1 – the opposite, and more degenerate, end of the same parameter range. So even where the sum-structure objection of (ii) did not apply, the parameter range needed for $A_{\pm}$ is exactly the range P1e already flags as needing new machinery, not an adaptation of what exists.

No zero of A_+A_- was found in a scan of 371 reduced $a/N \in (0, 0.0804)$, $3 \leq N \leq 151$ odd (P2v2_scan.py): the minimum $|A_+A_-|$ found was 0.1248 at $a/N = 6/115$, well away from 0. This is VERIFIED numerics only, consistent with non-vanishing on this finite range, and gives no route to a dense- N statement.

What would actually be needed. Per Section 4(ii), the natural next attempt is not a transplant of Theorem M but a direct estimate on the finite sum H_e (or H_o) as an incomplete quadratic Gauss sum with a slowly-varying ρ -dependent denominator weight $1/(\zeta; \zeta)_{2\rho}$: either (a) a stationary-phase/saddle treatment of the sum over ρ (which the main P2 note already does for the *outer* sum over n/k in B itself, suggesting the same q -Pochhammer-times-quadratic-phase technology might repackage H_e, H_o as a genuine (not block-model) saddle-point quantity, side-stepping non-vanishing of a literal finite sum in favour of non-vanishing of a leading saddle contribution), or (b) treating $A_{\pm}$ for a fixed small family of N (e.g. N in one residue class) as an exponential sum amenable to Weil/Weyl bounds. Neither is attempted here; both are OPEN and look like separate research atoms, not a corollary of Theorem M.

Weakest points of this note

- The renormalisation-limit check (Section 2) used a single seed $p/q = \frac{1}{13}$ and one approximating sequence ($a = \text{round}(N/13)$). Divergence there is strong evidence but not a proof that no renormalisation limit exists for *any* sequence or *any* seed; a cleverer choice of approximants (e.g. genuine continued-fraction convergents, or comparing $A_{\pm}(a/N)$ against $A_{\pm}$ at level q itself rather than against Z_q) was not tried.
- The term-spread check (Section 3) is a single case ($a/N \approx \frac{1}{13}$, $N = 131$). It was not repeated across the window or for H_o .
- The scan for zeros of A_+A_- (Section 5) only reaches $N \leq 151$; P2.note.tex’s own certified zeros go to $N = 17$ and its numerics to $N = 25$, so this is a modest extension in N but no closer to a density statement.
- No attempt was made at the alternative route (Section 5, “what would actually be needed”); this note only establishes that the *specific* technique named in the brief does not transfer, not that non-vanishing is intractable.
- The disanalogy (iii) (parameter-range mismatch with P1e’s golden threshold) was noted from reading P1e.lemma.tex, not independently re-derived; it is offered as a plausibility check on why (i)-(ii) are not a numerical accident.

Files. P2v2_scan.py (§5, the 371-case scan and the $|A_+A_-|$ extremes), P2v2_renorm.py (§2, the renormalisation-limit table), P2v2_terms.py (§3, the term-magnitude profile at $a = 10$, $N = 131$). All runs mpmath (Python), 30–40 decimal digits, each well under a second, run under `perl -e 'alarm`

290; `exec @ARGV'`. No BFS/enumeration; memory footprint negligible (a handful of 131×131 -scale complex arrays at 40 digits).

Room 3 (2026-09-26): two seats, moderator verdict **ADVANCED**

A room of mathematicians (two blind seats: Tao, Erdős; Reviewer AR independently rechecked the numerical finding) explored two further routes. Neither closes P2's accumulation question, but one produced a genuinely new asymptotic lead.

Tao's finding: the incomplete-Gauss-sum/stationary-phase route does not apply, and not for a generic reason. $H_e = \sum_{\rho} c^{\rho} \zeta^{\rho^2} / (\zeta; \zeta)_{2\rho}$ pairs a quadratic phase ζ^{ρ^2} with a weight $w_{\rho} = c^{\rho} / (\zeta; \zeta)_{2\rho}$. Every classical completion/van der Corput/stationary-phase-with-amplitude technique needs w_{ρ} to be a slowly-varying envelope. At $N = 131, a = 10$ it is not: $|w_{\rho}|$ spans four orders of magnitude and is non-monotone with multiple sign changes across ρ . w_{ρ} is itself a second oscillatory q -Pochhammer object, not a tame amplitude, so the hypothesis these methods share is genuinely violated, not merely untried.

Erdős's finding, confirmed by independent rerun (Reviewer AR): the ratio A_+/A_- converges to a root of unity along some (not all) seed families. Along $a/N \rightarrow 1/13$ ($a = \lfloor N/13 \rfloor$), $A_+/A_- \rightarrow -i$ (matched to 10^{-9} by $N = 2081$). Along $a/N \rightarrow 1/17$, it converges to $+i$ (confirmed at $N = 69633$: modulus and argument both correct to 8 displayed digits, though convergence is markedly slower than at $1/13$). Along $a/N \rightarrow 1/25$ it does *not* converge: $|A_+/A_-|$ climbs monotonically from 1.024 ($N = 25$) to 1.207 ($N = 102401$), confirmed not a rounding artifact (round vs. floor give identical residues throughout). This is a genuine asymptotic phenomenon, not a hidden exact identity: at finite N the ratio is never exactly modulus 1 (e.g. 1.126 at $N = 13$), only approaching it in the limit – new information not previously extracted from $A_{\pm}$'s separate behaviour.

What this is not, yet. No closed form for the limit (why $-i$ at $1/13$, $+i$ at $1/17$, divergence at $1/25$) was found; no mechanism connects this $N \rightarrow \infty$ -along-one-fixed-seed limit to the level- N -to-level- q induction Theorem M actually performs (this limit fixes the seed and lets N grow, not the other way around); and universality across seed families is unconfirmed (2 converge, 1 does not, out of 3 tested). A natural next step, not attempted: push $1/25$ much further, and attempt a closed form for the ratio via the same saddle-point/ q -Bessel technology B 's own asymptotics already use.

Room verdict: **ADVANCED.** A real, independently-reconfirmed new lead (the A_+/A_- ratio) alongside a precise negative diagnosis (the stationary-phase route's amplitude hypothesis fails mechanistically, not generically). P2's accumulation question stays OPEN; this is repo-only, not yet warranting a paper mention (no closed form, no induction mechanism, non-universal across the three seeds tried).